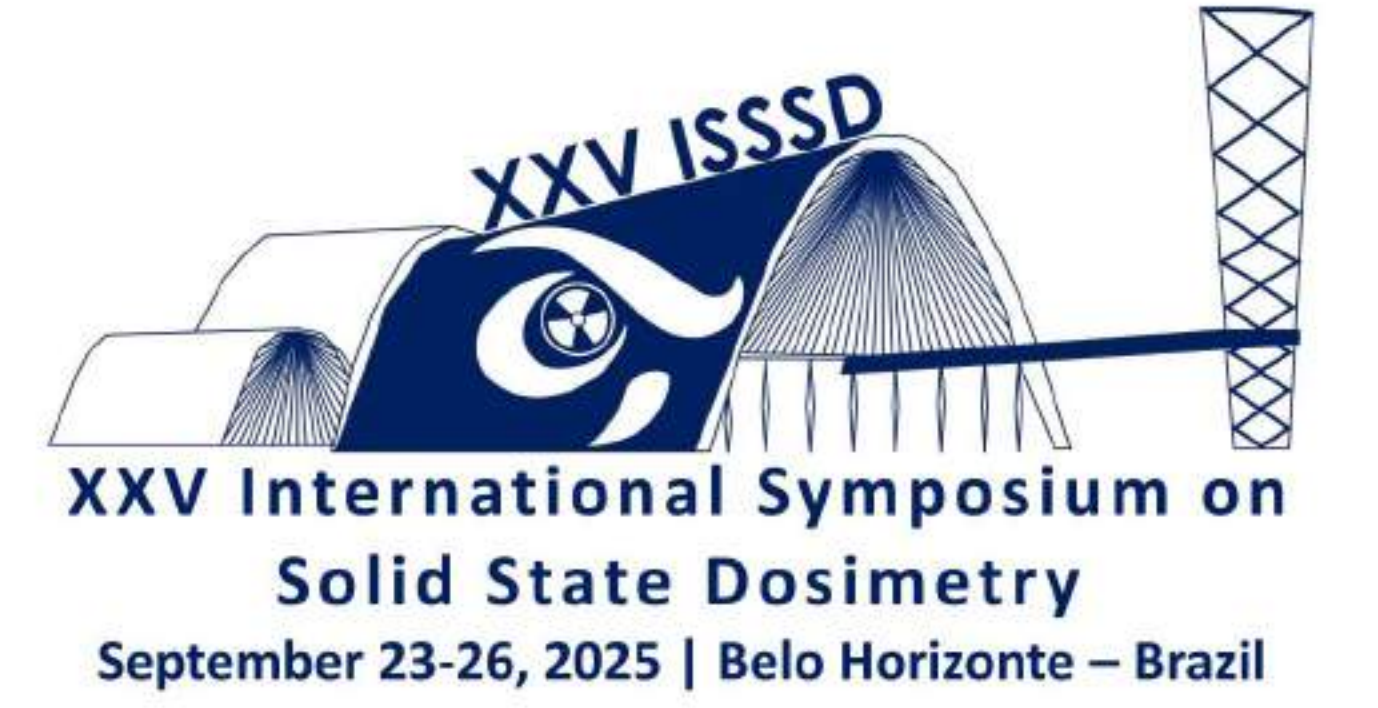


Physical and Computational Characterization of Electromagnetic Beam Intensity-Modulating Devices Using Monte Carlo Simulation

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1. INTRODUCTION

X-ray beams generated by bremsstrahlung in high-Z targets exhibit highly anisotropic fluence distributions, which constrains their use when specific or uniform profiles are required in scientific and biomedical applications. Modulation using physical devices, such as flattening filters, enables homogenization of the fluence within defined tolerances; however, the design details and parameters are often proprietary. In this work, we present a predictive applied-physics methodology, based on Monte Carlo simulations with PENELOPE together with theoretical considerations, to analyze the roles of geometry and material properties in achieving controlled and reproducible beam fluence distributions, including non-standard beam scenarios.

2. METHODS AND MATERIALS

A computational methodology based on Monte Carlo simulations with PENELOPE was implemented to model the primary component of bremsstrahlung X-ray beams and to systematically study the influence of the modulator geometry, atomic number, density, and linear attenuation coefficients. The flattening filter was selected as the beam modulator due to its historical role in fluence-profile homogenization and the available scientific evidence on its performance and design parameters.

The radial thickness profile, $x(r)$, was obtained by inverting the exponential attenuation law evaluated at the effective spectral energy of the photon spectrum, so that an incident profile, $I_0(r)$, is transformed into a target fluence, I_T .

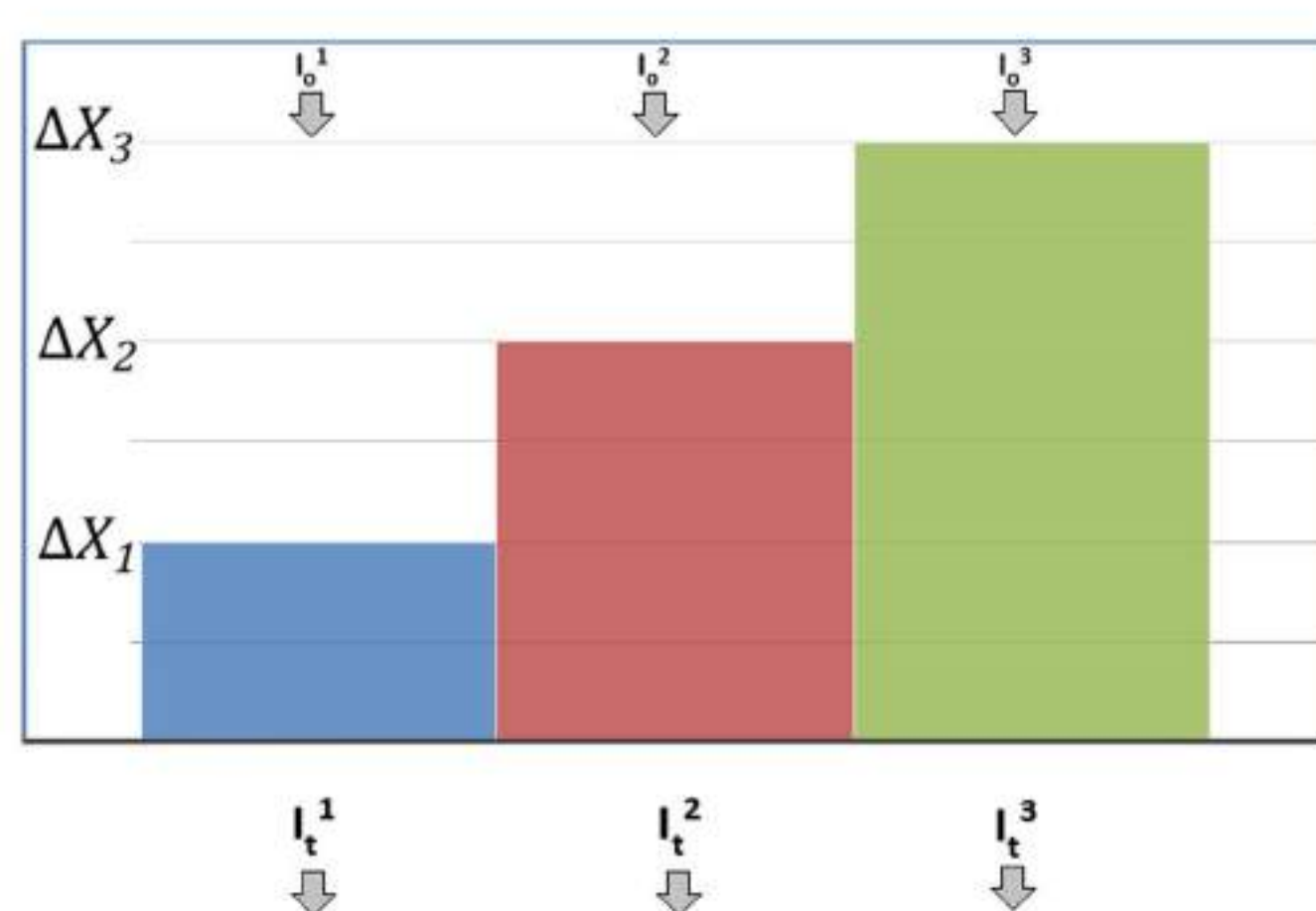


Fig. 1. The beam-fluence profile is discretized into bins. For each bin i with incident fluence $I_0^{(i)}$, the thickness increment ΔX_i is estimated so

that the output fluence $I_t^{(i)}$ matches the target I_T .

Consequently, $\Delta X_3 > \Delta X_2 > \Delta X_1$ if $I_0^{(3)} > I_0^{(2)} > I_0^{(1)}$.

$$x(r) = \frac{1}{\mu_l(E_{ef})} \ln \left[\frac{I_0(r)}{I_T} \right]$$

$\mu_l(E_{ef})$ denotes the linear attenuation coefficient evaluated at the effective spectral energy.

Under this formulation, the inverted-attenuation approach yields physically plausible and reproducible modulator designs for multiple materials (tungsten, copper, aluminum) and remains applicable to non-standard beam conditions.

3. RESULTS

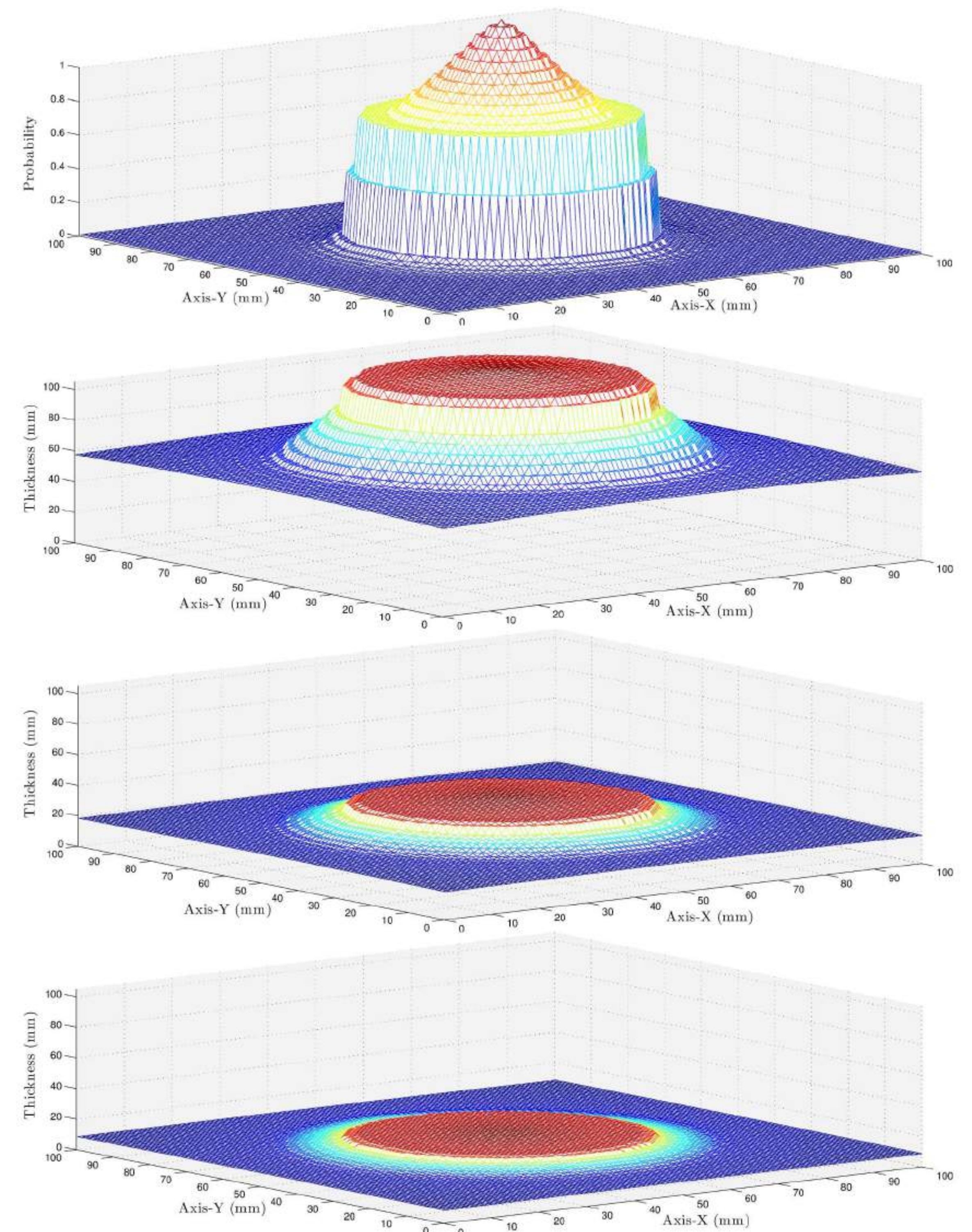


Fig. 2. The top panel shows the incident fluence field $I_0(x, y)$ obtained by Monte Carlo simulation (PENELOPE). The bottom panels display the thickness maps for Al, Cu, and W, respectively, computed by local inversion of the exponential attenuation using the linear attenuation coefficient $\mu_l(E_{ef})$.

With the same incident photon field, generated by bremsstrahlung from 6 MeV electrons, and a common target fluence I_T , flattening filters for W, Cu, and Al were obtained by local inversion of the attenuation law with $\mu_l(E_{ef})$. Although the thickness maps differ, all designs produce the same transmitted fluence field $I_T(x, y)$. The thickness magnitudes follow the expected trends with atomic number and density.

4. CONCLUSIONS

The results validate the proposed methodology. In the design of flattening filters, a historical fluence-modulation device, the thickness maps obtained for W, Cu, and Al were consistent with practice for the same target fluence. The approach is general and can be applied to other beam qualities and energies in biomedical, industrial, and applied-physics research and applications.

REFERENCES

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